# Firm Entry and AI: Evidence from the US

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Preliminary and incomplete. Comments welcome.

**Abstract**

[Abstract placeholder. Summarise: motivation, data, method, main finding, contribution. Roughly 150–200 words.]

**Keywords:** business formation, artificial intelligence, AI exposure, difference-in-differences, ChatGPT

**JEL codes:** L26, O33, J24, M13

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# 1 Introduction

The arrival of large language models—most visibly the public release of ChatGPT in November 2022—has revived interest in how general-purpose technologies reshape labour markets and firm dynamics (Felten et al., 2021, Fossen et al., 2024). Most of this debate focuses on *workers*: which jobs are at risk, and for whom. But a parallel question may be equally consequential: does AI alter the incentives to *start* a business?

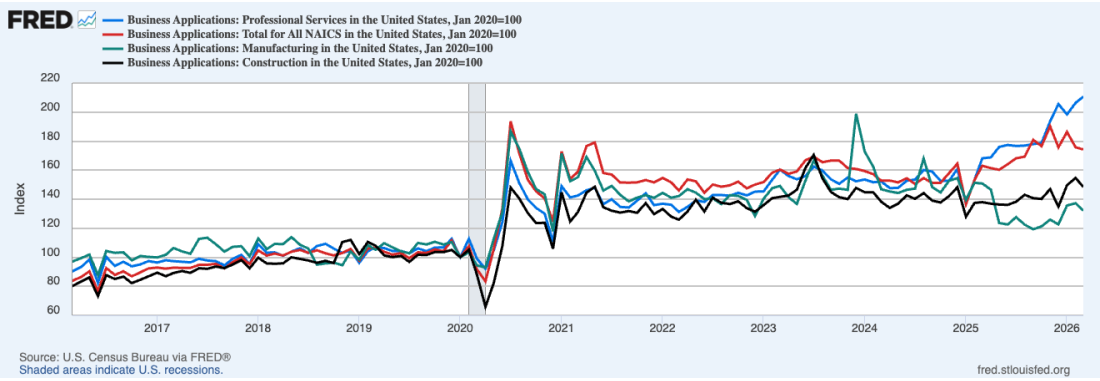


Figure 1: Business applications indexed to January 2020 = 100 for selected sectors. Source: U.S. Census Bureau via FRED (https://fred.stlouisfed.org/graph/?g=1UHD5). Shaded area indicates the COVID-19 recession (Feb–Apr 2020).

Figure 1 offers a first look. After the COVID-19 shock, aggregate business applications surged to historic highs and have continued rising through 2025. But the expansion is not uniform. Professional and Business Services—among the most AI-exposed sectors by any measure—have pulled sharply away from the aggregate trend. Manufacturing and Construction, where tasks are less amenable to language-model automation, have lagged behind. This heterogeneity motivates the present paper.

Lower entry costs, improved productivity for high-skilled tasks, and easier access to specialised knowledge may all reduce barriers to entrepreneurship in AI-exposed sectors (Brynjolfsson et al., 2018). Conversely, if AI primarily benefits large incumbents, competitive pressures could *raise* barriers to entry. Distinguishing between these channels requires variation both across sectors and over time—variation the public release of ChatGPT conveniently provides.

We use monthly sector-level business application counts from the Census Bureau Business Formation Statistics (Haltiwanger et al., 2020) together with the AI Industry Exposure (AIIE) index of Felten et al. (2021) to estimate a difference-in-differences (DiD) model of the form

$$\log(\mathrm{BA}_{it}) \;=\; \alpha_i + \gamma_t + \beta\,(\mathrm{Post}_t \times \text{AI Exposure}_i) + \varepsilon_{it}, \tag{1}$$

where $i$ indexes 2-digit NAICS sectors, $t$ indexes calendar months, $\mathrm{Post}_t$ is an indicator equal to one from November 2022 onwards, $\alpha_i$ are sector fixed effects absorbing time-invariant differences in application rates, $\gamma_t$ are month fixed effects absorbing aggregate cyclical variation, and standard errors are clustered by sector. The coefficient $\beta$ identifies whether sectors with higher AI exposure experienced a differential change in business formation after ChatGPT.

**Related literature.** Fossen et al. (2024) examine AI adoption and self-employment at the individual level and find positive effects concentrated among high-skilled workers. Chatterji et al. (2025) provide a conceptual treatment of how transformative AI reshapes firm organisation, adoption incentives, and competitive dynamics—a natural complement to the reduced-form evidence presented here. Felten et al. (2021) construct the AIOE/AIIE index by combining O*NET task measures with AI capability scores. Haltiwanger et al. (2020) describe the Census Bureau BFS and demonstrate its value for studying business dynamism and entry margins. Brynjolfsson et al. (2018) provide a conceptual framework for mapping task-level AI capabilities to occupational and industry outcomes.

# 2 Data

## 2.1 Business Formation Statistics

The Census Bureau BFS provides monthly, seasonally adjusted counts of new Employer Identification Number (EIN) applications at the 2-digit NAICS level, derived from IRS Form SS-4 filings (Haltiwanger et al., 2020). EIN applications are a leading indicator of actual busi-

ness formation: roughly half of high-propensity applicants open within eight quarters of filing. We use the official seasonally adjusted series directly and index each sector to January 2020 = 100, yielding a panel observed from January 2006 through early 2026. The BFS publishes 20 sector-level series (excluding the unclassified "No NAICS" category): Manufacturing (NAICS 31–33), Retail Trade (44–45), and Transportation & Warehousing (48–49) are each published as single combined codes rather than individual 2-digit breakdowns. Public Administration (NAICS 92) is present in the BFS but has no corresponding AIOE score and is therefore excluded from the regression sample, leaving 19 sectors. We define two treatment dates: November 2022 (public launch of ChatGPT) and May 2025 (widespread availability of agentic AI tools).

Figure 2 plots the index for four selected sectors. The divergence between Professional Services and goods-producing sectors is visible from approximately 2021 and widens after November 2022, consistent with a role for AI in lowering barriers to entry specifically where tasks are more cognitively intensive.

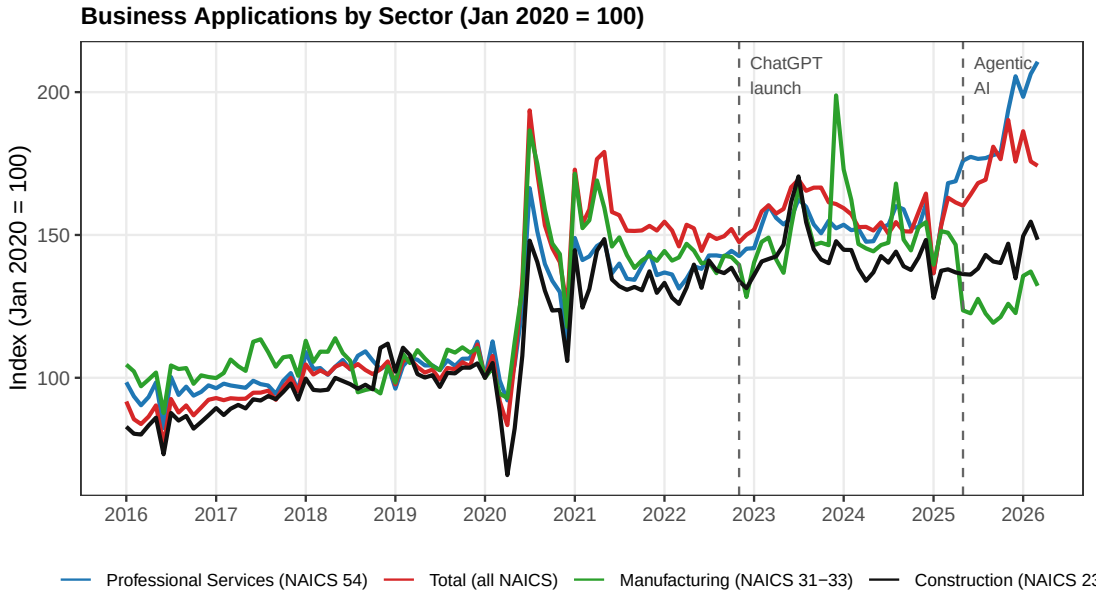


Figure 2: Business applications indexed to January 2020 = 100. Professional Services (NAICS 54), Total (all NAICS), Manufacturing (NAICS 31–33), and Construction (NAICS 23). Dashed vertical lines mark November 2022 (ChatGPT launch) and May 2025 (agentic AI tools). Seasonally adjusted. Source: Census Bureau BFS.

## 2.2 AI Exposure

Following Felten et al. (2021), we use the AI Industry Exposure (AIIE) index, which maps occupational AI exposure scores to 4-digit NAICS industries through BLS employment weights and aggregates to 2-digit NAICS by employment-weighted averaging. The index is constructed from 10 AI application domains benchmarked against O*NET task descriptions and captures the degree to which a sector's occupational mix is susceptible to AI automation or augmentation. Finance and Insurance (NAICS 52, AIIE = 2.05), Professional Services (NAICS 54, AIIE = 1.59), and Information (NAICS 51, AIIE = 1.27) rank highest; Agriculture (NAICS 11, AIIE = −1.56), Transportation (NAICS 49, AIIE = −1.31), and Construction (NAICS 23, AIIE = −1.00) rank lowest. This cross-sectional variation—which is fixed before the ChatGPT shock—is the key source of identification. Table 1 reports the AIIE score alongside summary statistics of the BFS series for each sector in the sample.

Table 1: AI Industry Exposure and Business Formation Statistics by Sector

| Sector | NAICS | AIIE | Mean | SD | Min | Max | $N$ |
|---|---|---|---|---|---|---|---|
| | | | Monthly BA (thousands) | | | | (months) |
| Finance & Insurance | 52 | 2.05 | 14.0 | 3.6 | 8.4 | 22.0 | 261 |
| Management of Companies | 55 | 1.70 | 2.2 | 1.3 | 0.9 | 6.6 | 261 |
| Professional, Scientific, Technical Svcs | 54 | 1.59 | 38.2 | 11.7 | 19.8 | 76.7 | 261 |
| Educational Services | 61 | 1.32 | 4.6 | 1.5 | 2.4 | 8.6 | 261 |
| Information | 51 | 1.27 | 6.1 | 1.8 | 3.0 | 11.4 | 261 |
| Health Care & Social Assistance | 62 | 0.56 | 18.3 | 5.6 | 10.2 | 33.1 | 261 |
| Real Estate & Rental | 53 | 0.51 | 16.9 | 5.4 | 8.9 | 26.5 | 261 |
| Wholesale Trade | 42 | 0.26 | 8.9 | 1.3 | 6.1 | 16.2 | 261 |
| Utilities | 22 | 0.13 | 0.3 | 0.2 | 0.1 | 0.8 | 261 |
| Arts, Entertainment, & Recreation | 71 | 0.10 | 9.5 | 2.2 | 5.3 | 15.4 | 261 |
| Retail Trade (NAICS 44–45) | RET | -0.01 | 45.2 | 24.7 | 20.1 | 131.0 | 261 |
| Other Services | 81 | -0.03 | 26.6 | 10.0 | 12.7 | 62.8 | 261 |
| Admin. & Support Services | 56 | -0.21 | 20.3 | 7.2 | 10.9 | 41.9 | 261 |
| Manufacturing (NAICS 31–33) | MNF | -0.57 | 5.7 | 1.3 | 3.7 | 10.5 | 261 |
| Mining, Quarrying, Oil & Gas | 21 | -0.68 | 0.6 | 0.1 | 0.3 | 1.4 | 261 |
| Transportation & Warehousing (48–49) | TW | -0.77 | 18.8 | 9.7 | 7.9 | 44.0 | 261 |
| Accommodation & Food Services | 72 | -0.93 | 15.6 | 5.5 | 8.5 | 31.6 | 261 |
| Construction | 23 | -1.00 | 29.7 | 9.1 | 16.0 | 52.9 | 261 |
| Agriculture, Forestry, Fishing | 11 | -1.56 | 3.5 | 1.5 | 1.9 | 19.4 | 261 |

*Notes:* AIIE = AI Industry Exposure index from Felten et al. (2021), constructed from O*NET task descriptions and aggregated from 4-digit to 2-digit NAICS by employment-weighted averaging. Mean, SD, Min, and Max refer to monthly seasonally adjusted business applications (thousands) over the full sample period (January 2006 to early 2026). $N$ is the number of monthly observations. Sectors sorted by AIIE score descending.

# 3 Empirical Strategy

[Empirical strategy placeholder.]

Our baseline specification is equation (1). The identifying assumption is that, absent the ChatGPT launch, business applications would have evolved in parallel across sectors regardless of their AI exposure. We probe this assumption with an event-study specification that replaces $\text{Post}_t$ with a full set of month-relative-to-November-2022 indicators interacted with $\text{AI Exposure}_i$. Parallel pre-trends would be consistent with the identification assumption.

Standard errors are clustered at the 2-digit NAICS sector level using `fixest` (Bergé, 2018).

We probe robustness along two dimensions. First, we substitute the outcome variable with the full suite of BFS series, which span the entire formation funnel. On the applications side: corrected business applications (CBA), which purge filings unlikely to yield an employer business; high-propensity applications (HBA); and applications with planned wages (WBA). On the formations side: businesses actually formed within four quarters (BF4Q) and eight quarters (BF8Q) of filing; Census-projected formations at both horizons (PBF4Q, PBF8Q); and seasonally-adjusted formation counts (SBF4Q, SBF8Q). Finally, the median number of weeks from application to formation (DUR4Q, DUR8Q) provides a complementary timing dimension: a negative coefficient would indicate that AI exposure compresses the application-to-opening lag, consistent with lower operational setup costs. If AI lowers barriers to entry, the effect should be visible across all stages of the formation funnel. Second, we re-estimate the model at the state level, replacing sector fixed effects with state fixed effects. The outcome is the log of total seasonally adjusted business applications aggregated across all sectors within a state, and the AI exposure measure is the AI Geographic Exposure (AIGE) index from Felten et al. (2021) Appendix C, which is constructed directly from the occupational mix of each state and its workers' AI exposure. This specification absorbs permanent state-level differences in entrepreneurial activity and exploits cross-state variation in AI exposure that is independent of the sector composition used in the main analysis.

# 4 Results

## 4.1 Descriptive Evidence

Figure 3 plots mean indexed business applications by AI exposure quartile. Prior to 2020 the four quartile groups track closely together, consistent with parallel pre-trends. After the COVID-19 surge and recovery, the top quartile (high AI exposure) diverges upward, while the bottom quartile remains below the aggregate trend. The gap is most pronounced from 2024 onwards, consistent with a gradual diffusion of AI tools into entrepreneurial activity rather than an immediate discrete response.

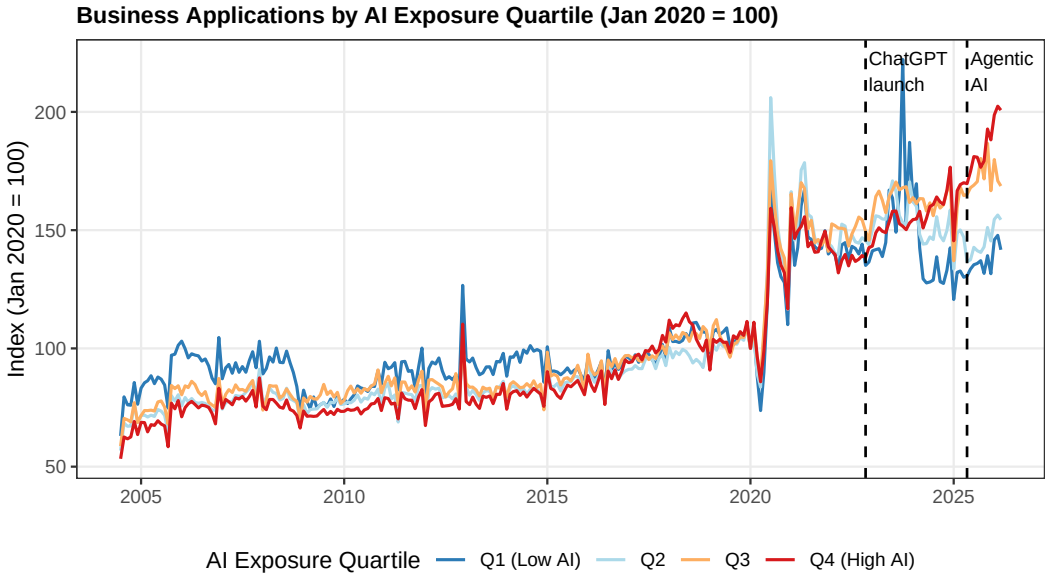


Figure 3: Business applications by AI exposure quartile (Jan 2020 = 100). Dashed vertical lines mark November 2022 (ChatGPT launch) and May 2025 (agentic AI tools).

## 4.2 Difference-in-Differences

Table 2 reports the DiD estimates. Columns (1)–(3) pool the entire post-November 2022 period into a single *Post* indicator; column (4) splits the post period into a ChatGPT window (November 2022–April 2025) and an agentic-AI window (May 2025 onwards). The single-period estimates are attenuated relative to the two-period specification, consistent with the

Table 2: Difference-in-Differences Estimates

| | (1) log(BA) | (2) BA Index | (3) BA Level | (4) Two-period |
|---|---|---|---|---|
| $\text{Post}_{\text{ChatGPT}} \times$ AI Exposure | 0.209 | 30.151 | -1607.456 | 0.121 |
| | (0.227) | (30.712) | (7157.107) | (0.214) |
| Observations | 4959 | 4959 | 4959 | 4959 |
| $R^2$ | 0.982 | 0.690 | 0.834 | 0.983 |
| Sector FE | ✓ | ✓ | ✓ | ✓ |
| Month FE | ✓ | ✓ | ✓ | ✓ |

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by 2-digit NAICS sector. $\text{Post}_{\text{ChatGPT}}$: Nov 2022 – Apr 2025. $\text{Post}_{\text{Agentic}}$: May 2025 onwards.

event-study evidence that the effect builds gradually and becomes strongest toward the end of the sample.

## 4.3 Event Study

Figure 4 presents the event-study coefficients. The pre-period estimates ($t < 0$) are small and statistically indistinguishable from zero, supporting the parallel-trends assumption. Post-launch, coefficients rise gradually before the 95 % confidence interval clears zero around 2025 ($t \approx +25$ to $+30$), indicating that the differential effect of AI exposure on business formation became statistically significant roughly two to three years after the ChatGPT launch. This pattern is consistent with a diffusion mechanism: entrepreneurs require time to learn the tools, identify viable business models, and translate that into actual EIN filings. A visible second inflection around $t = +30$ (May 2025, the emergence of widely available agentic AI tools) suggests a possible amplification of the initial shock, which the two-period DiD in column (4) of Table 2 is designed to capture.

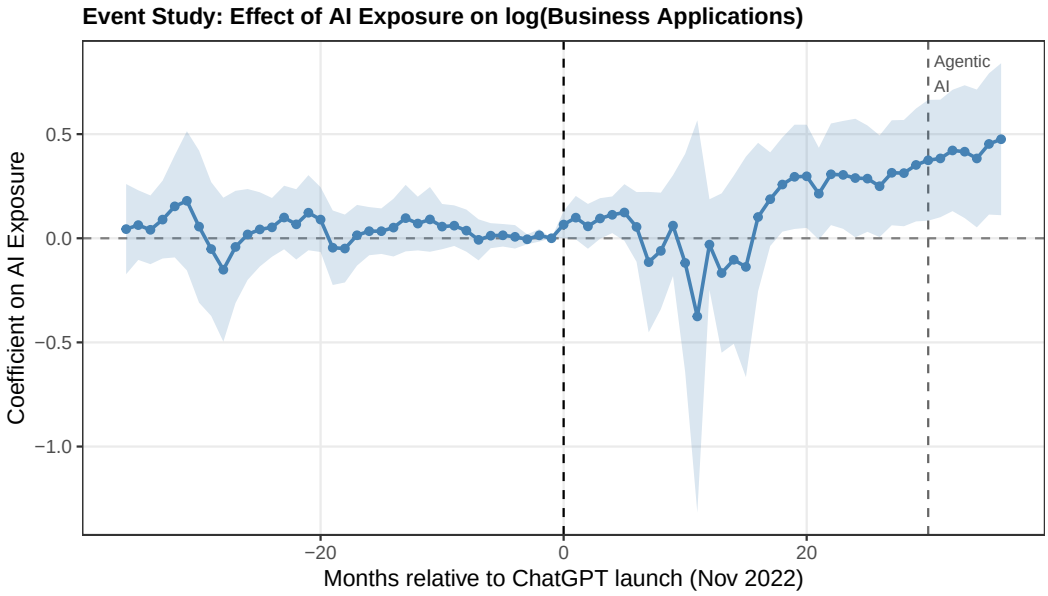


Figure 4: Event-study estimates: effect of AI exposure on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on sector-clustered standard errors. The second dashed line marks May 2025 ($t = +30$, agentic AI).

## 4.4 Robustness Checks

### Alternative BFS series

Table 3 replicates the baseline DiD specification (column (1) of Table 2) using three alternative BFS outcome variables. Column (2) replaces total business applications with high-propensity applications (HBA), defined by the Census Bureau as those most likely to become active employer businesses. Column (3) uses applications with planned wages (WBA), which signal explicit intent to hire at the time of EIN filing. Column (4) uses the count of businesses that actually formed within four quarters of application (BF4Q), the realised formation margin. A consistent pattern across the formation funnel—from intent (WBA) to high-quality applications (HBA) to actual formations (BF4Q)—strengthens the interpretation that AI exposure shifted both the quantity and quality of entrepreneurial activity, not merely the propensity to file. We additionally examine formations within eight quarters (BF8Q) in the event-study figures; because the Census requires a full eight-quarter follow-up window

before publishing BF8Q, the series is right-truncated approximately two years before the data vintage and does not have sufficient post-period coverage for a clean DiD estimate. The event-study coefficients are nevertheless informative: comparing the BF4Q and BF8Q panels reveals whether the additional entry in high-AI-exposure sectors converts to operating businesses quickly or gradually.

Figure 5 presents the corresponding event-study coefficients for all four series. The pre-period estimates are small and insignificant across every panel, ruling out differential pre-trends. Post-launch dynamics are qualitatively consistent: coefficients drift upward at similar timing and reach statistical significance in the same window regardless of which formation margin is used as the outcome. The BF4Q panel is shifted rightward relative to applications panels, reflecting the mechanical lag between filing and actual business opening, and provides perhaps the cleanest evidence that the effect translates into realised entry rather than merely intent.

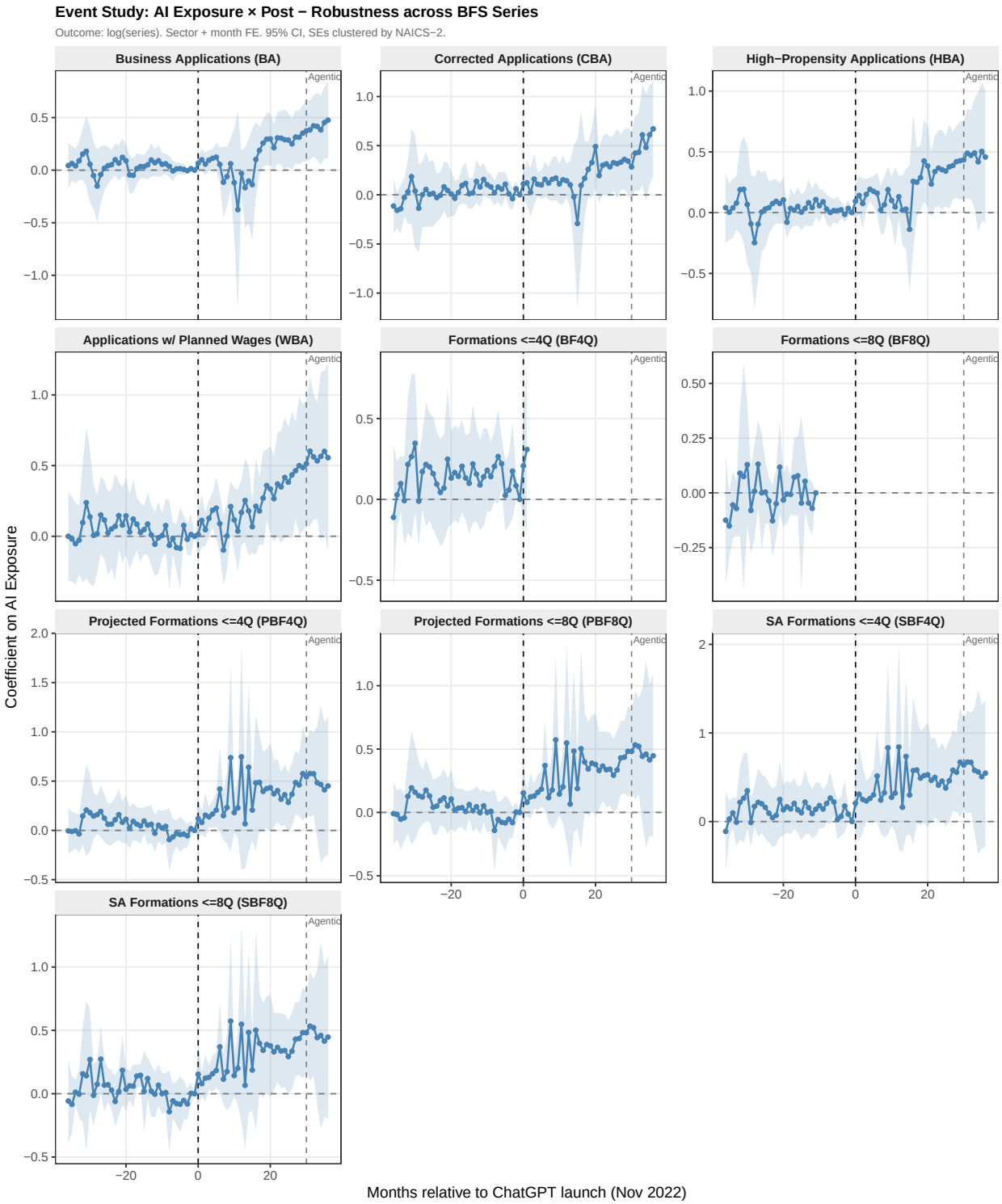


Figure 5: Event-study estimates for four BFS series. Each panel plots the interaction coefficient between AI exposure and a month-relative-to-November-2022 indicator. BA = total applications; CBA = corrected applications; HBA = high-propensity applications; WBA = applications with planned wages; BF4Q/BF8Q = formations within 4/8 quarters; PBF4Q/PBF8Q = projected formations; SBF4Q/SBF8Q = seasonally-adjusted formations; DUR4Q/DUR8Q = median weeks to formation (levels, not logs). Shaded bands are 95 % confidence intervals based on sector-clustered standard errors. All specifications include sector and month fixed effects. Dashed vertical lines mark November 2022 ($t = 0$) and May 2025 ($t = +30$, agentic AI).

Table 3: Robustness: Alternative BFS Series

| | (1) log(BA) | (2) log(CBA) | (3) log(HBA) | (4) log(WBA) | (5) log(BF4Q) |
|---|---|---|---|---|---|
| Post × AI Exposure | 0.209 | 0.426 | 0.212 | 0.287 | 0.377 |
| | (0.227) | (0.292) | (0.291) | (0.267) | (0.283) |
| Observations | 4959 | 4959 | 4959 | 4959 | 4216 |
| $R^2$ | 0.982 | 0.980 | 0.982 | 0.984 | 0.988 |
| Sector FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| Month FE | ✓ | ✓ | ✓ | ✓ | ✓ |

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01

Standard errors clustered by 2-digit NAICS sector. All specifications include sector and month fixed effects. BA = total business applications; HBA = high-propensity business applications; WBA = applications with planned wages; BF4Q = businesses formed within four quarters of application.

### State-level panel with state fixed effects

Table 4 presents the state-level DiD estimates. The unit of observation is now a state-month cell, with the outcome being log total seasonally adjusted business applications aggregated across all sectors within the state. The AI exposure measure is the AIGE index of Felten et al. (2021), which captures the AI exposure of a state's workforce through its occupational mix. State fixed effects absorb permanent cross-state differences in entrepreneurial culture, regulatory environment, and industrial composition; month fixed effects absorb national macro shocks as before. Standard errors are clustered at the state level (56 clusters).

Figure 6 provides raw descriptive evidence at the state level, analogous to Figure 3 for sectors. States are sorted into quartiles by their AIGE index and the cross-state mean of indexed business applications (January 2020 = 100) is plotted for each quartile.

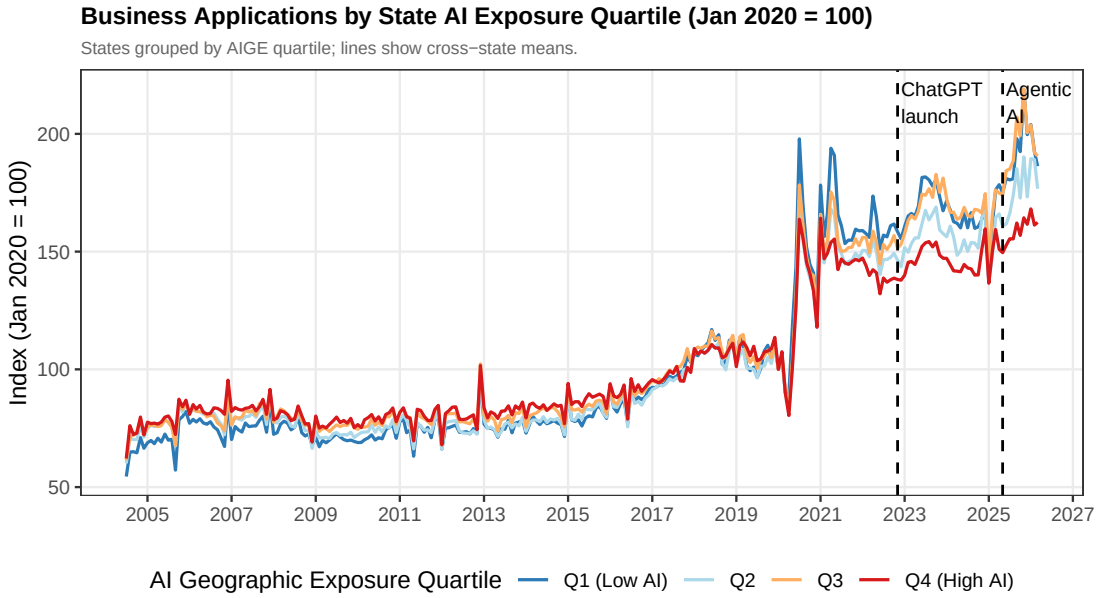


Figure 6: Mean business applications (index, January 2020 = 100) by AI Geographic Exposure (AIGE) quartile across US states and territories. Q1 denotes the lowest-exposure quartile; Q4 the highest. Lines show simple cross-state means within each quartile. Dashed vertical lines mark November 2022 (ChatGPT launch) and May 2025 (agentic AI wave).

Column (1) shows the single-period estimate; column (2) uses the indexed BA outcome; column (3) splits the post period as in the main analysis. The state-level estimates are

Table 4: State-Level Difference-in-Differences Estimates

| | (1) log(BA) | (2) BA Index | (3) Two-period |
|---|---|---|---|
| $\text{Post}_{\text{ChatGPT}} \times$ AI Exposure | -0.445 | -67.191 | -0.423 |
| | (0.274) | (42.331) | (0.265) |
| Observations | 13572 | 13572 | 13572 |
| $R^2$ | 0.986 | 0.825 | 0.986 |
| State FE | ✓ | ✓ | ✓ |
| Month FE | ✓ | ✓ | ✓ |

* p ¡ 0.1, ** p ¡ 0.05, *** p ¡ 0.01
Standard errors clustered by state. AI exposure is the AI Geographic Exposure (AIGE) index from Felten et al. (2021) Appendix C, constructed from occupation-level AIOE scores weighted by the state employment-mix across occupations. $\text{Post}_{\text{ChatGPT}}$: Nov 2022 – Apr 2025. $\text{Post}_{\text{Agentic}}$: May 2025 onwards.

statistically insignificant across all specifications, in contrast to the sector-level results in Table 2.

Figure 7 shows the corresponding event-study coefficients, and Figure 8 repeats the exercise across the four BFS series published at the state level (BA, HBA, WBA, CBA; formation series are national-only). Pre-period coefficients are flat across all panels, consistent with parallel pre-trends, but the post-launch coefficients are imprecise and do not reach conventional significance levels in any series.

The absence of a detectable state-level effect likely reflects two compounding limitations. First, states are highly diversified: even a state with a high-AIGE workforce will contain many low-AI sectors that dilute the aggregate application signal. The sector-level design exploits within-sector variation in AI exposure that is averaged away when outcomes are aggregated to the state level. Second, the cross-state dispersion of AIGE is considerably smaller than the cross-sector dispersion of AIOE, reducing statistical power. Together, these features make the state-level panel a demanding test; the sector-level estimates remain the primary evidence.

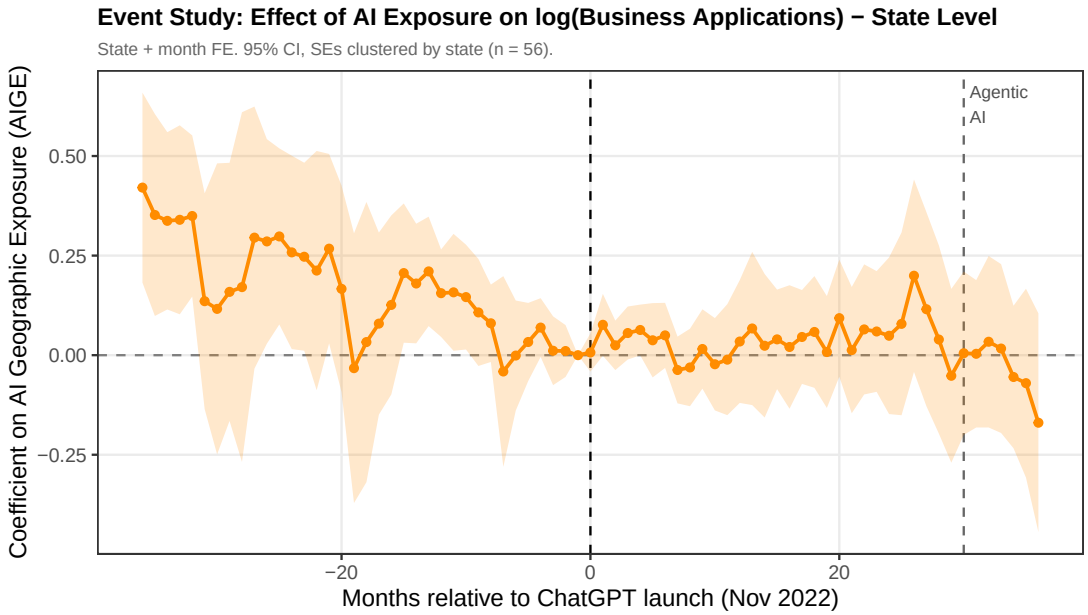


Figure 7: State-level event-study estimates: effect of AI Geographic Exposure (AIGE) on log(Business Applications) relative to November 2022 ($t = 0$). Shaded band is the 95 % confidence interval based on state-clustered standard errors (56 clusters). State and month fixed effects included. The second dashed line marks May 2025 ($t = +30$, agentic AI). Coefficients are statistically insignificant throughout.

Figure 8 replicates the state-level event study across the four BFS series available at state level.

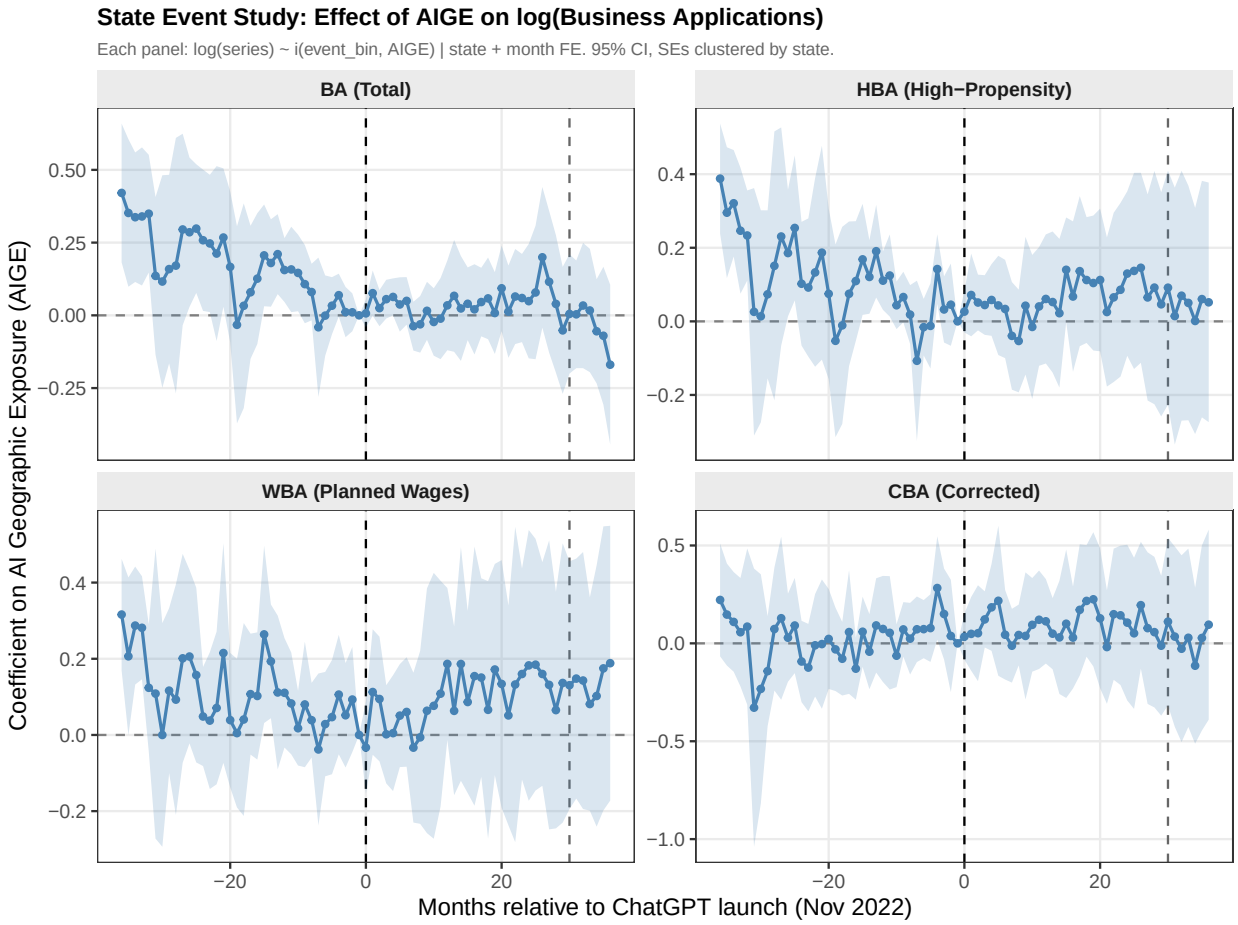


Figure 8: State-level event-study estimates for four BFS series. Each panel plots the interaction coefficient between AI Geographic Exposure (AIGE) and a month-relative-to-November-2022 indicator, estimated on the state × month panel. BA = total applications; HBA = high-propensity applications; WBA = applications with planned wages; CBA = corrected applications. Business formation series (BF4Q, BF8Q, etc.) are not published at state level by the Census Bureau. Shaded bands are 95 % confidence intervals based on state-clustered standard errors (56 clusters). All specifications include state and month fixed effects. Dashed vertical lines mark November 2022 ($t = 0$) and May 2025 ($t = +30$, agentic AI).

# 5 Conclusion

[Conclusion placeholder.]

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